

Session 2 Lab Exercise

For the lab exercise, we'll use a dataset related to crime rates in different U.S. states. The dataset includes variables such as the crime rate, urbanization rate, and average length of education. Explore the relationships between these variables using covariance, correlation, and OLS regression.

```
# Load the necessary package
library(MASS)
# Load the USArrests dataset
data(USArrests)
```

Tasks

1. Calculate the covariance between murder and urban population.
2. Calculate the correlation between murder and urban population.
3. Perform OLS regression with murder rate as the dependent variable
4. Interpret the coefficients
5. What is the R-squared and interpret it?
6. Create a scatter plot with the fitted regression line
7. This next task will require you to use the following dataset: day1.csv.

Ignoring what the variables are for a moment, estimate the following 4 linear models using OLS and report the results:

```
model1 <- lm(y1 ~ x1, data)
model2 <- lm(y2 ~ x2, data)
model3 <- lm(y3 ~ x3, data)
model4 <- lm(y4 ~ x4, data)
```

8. Now implement the judgement of a good social scientist. How would you compare the linear models you just estimated?